

Uladzislau Rybakou

Backend / Full Stack Engineer

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Profile

Backend-focused Software Engineer with 5+ years of experience specializing in Node.js (TypeScript, NestJS). Strong track record of designing scalable APIs, handling complex database architectures (PostgreSQL), and building robust cross-system integrations. Experienced in safely managing data migrations, modernizing legacy codebases, and leveraging advanced AI-driven development workflows (Claude, Antigravity, Cursor) to drastically accelerate time-to-delivery. Alongside commercial engineering, creator of several independent products — [see Featured Projects below](#) ↓. Available for a full-time commitment.

Technical Skills

Languages: TypeScript JavaScript SQL

Backend Core: Node.js NestJS Express REST APIs WebSockets

Data & Queues: PostgreSQL pgvector ClickHouse Kafka BullMQ Redis

AI & Workflow: AI Agents Multimodal AI (VLM) Claude Antigravity Custom Skills & Hooks Cursor

DevOps & Testing: AWS (RDS, SQS, ECS, Lambda) Cloudflare Docker CI/CD (GitHub Actions, GitLab CI) Jest Git

Employment History

- Backend Developer | Blask, Tbilisi

Jul 2024 — Jun 2026

 - Database Engineering & Migrations:** Managed complex PostgreSQL schemas, optimized analytical queries, and executed secure database updates to maintain high data consistency across microservices.
 - System Integrations:** Engineered resilient cross-system integration pipelines using Kafka, BullMQ, and Redis, preventing data loss through Dead Letter Queues (DLQ) and retry mechanisms.
 - Architecture & Testing:** Developed scalable REST APIs with NestJS and TypeScript. Maintained high system reliability by covering critical integration points and business logic with comprehensive unit and E2E tests (Jest).
 - AI-Driven Development:** Utilized advanced AI orchestration tools and agentic workflows (Antigravity IDE, Claude) to rapidly analyze infrastructure, fast-track legacy refactoring, and automate boilerplate code extraction.
- Full Stack Developer | DIBSCapital, Tel Aviv

Jun 2022 — Jun 2024

 - Third-Party Integrations:** Connected internal core systems with external partner applications (North Capital KYC, InvestReady, Stripe), facilitating secure user data flow and synchronization.
 - Cloud Infrastructure:** Deployed and managed backend services within AWS environments (ECS, Lambda, RDS), scaling infrastructure to support sensitive transactions reliably.
 - Feature Delivery:** Developed and optimized robust full-stack solutions, ensuring seamless integration between Node.js backend services and React frontend interfaces.
- Backend Developer | Nublecraft

Apr 2021 — Jun 2022

 - System Modernization:** Audited and iteratively refactored legacy codebases to modernize APIs and improve overall system performance without disrupting active business operations.
 - API Development:** Built, optimized, and documented reliable server-side REST APIs using Node.js for seamless integration by internal and external clients.
- Live Video Streaming Engineer | Specmedia, Minsk

Mar 2017 — Jun 2020

 - Infrastructure Management:** Maintained live streaming infrastructure, ensuring continuous uptime, network optimization, and stable video delivery pipelines.

Featured Projects

Stsezhka | stsezhka.com

Active Project

Full-scale outdoor routing and mapping platform designed for cycling, running, and hiking adventures, featuring an interactive mapping environment and custom-built routing services.

- **Custom Routing & Geospatial Engine:** Engineered multi-modal route planning with elevation profiling, gradient analysis, surface classification, and automated cue sheet / roadbook generation.
- **High-Performance Architecture:** Built within a TypeScript monorepo (Turborepo/pnpm), processing complex GPX datasets, waypoint sequencing, and smooth map interactions (Mapbox / OSM).
- **Scalable Ecosystem:** Designed a resilient backend API alongside an offline-first web application and cross-platform mobile client.

TypeScript Node.js Turborepo Mapbox / OSM Geospatial / GPX REST API

SplitBot | botsplit.com

Production / Live

Production-grade Telegram Mini App and bot for collaborative group expense splitting with zero-friction onboarding and multimodal receipt processing.

- **Multimodal AI Receipt OCR:** Built an automated recognition pipeline using Vision-Language Models (VLM) that extracts line items from receipt photos and translates international dish names in a single inference pass.
- **Real-Time Synchronization:** Implemented collaborative item claiming via WebSockets and Telegram HMAC authentication, enabling group members to split bills simultaneously with instant state sync.
- **Financial Integrity & Algorithms:** Implemented an automated debt-simplification algorithm that minimizes the number of settlement transfers between participants; multi-currency conversions and proportional tax/tip allocations.
- **Global Localization:** Localized across 21 languages, deployed on Cloudflare Pages with Dockerized backend infrastructure.

TypeScript Next.js Python FastAPI PostgreSQL Redis WebSockets Multimodal AI (VLM) Telegram Mini App Cloudflare

NeyroNews | neyronews.com

Production / Live

Autonomous multi-agent news curation and broadcasting pipeline delivering technical AI & developer news with Human-in-the-Loop moderation.

- **Semantic Deduplication & Crawling:** Continuously monitors multi-channel web & RSS sources; eliminates duplicates using exact SHA-256 hashing and semantic cosine similarity with PostgreSQL `pgvector`.
- **Two-Stage LLM Evaluation Funnel:** Implemented a tiered scoring architecture — high-throughput lightweight LLM for initial triage, followed by advanced LLM synthesis to draft publication-ready technical posts.
- **HITL Moderation & Concurrency:** Telegram inline callback buttons for editor consensus review and a scheduled publication engine using PostgreSQL `SELECT ... FOR UPDATE SKIP LOCKED` to prevent race conditions.

Python PostgreSQL pgvector LiteLLM LLM Pipelines Telegram Bot API Docker